Breaking Black Box - Evals

Craig West

https://craig-west.netlify.app/

https://evaluating-ai-agents.com/

- Contains links for slides and eval-framework
- Everything in this talk is included

My desired outcome

Not for you to fully understand Evals

But…

That Evals are doable

There are useful resources available for you

Demystify and simplify what an Agent is

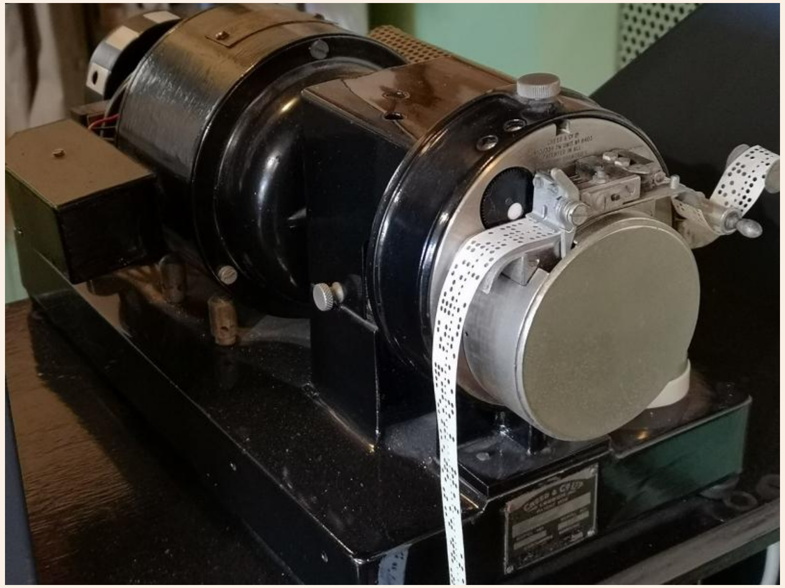
There will be some slides and then we will dive into the code and manual (website).

About me

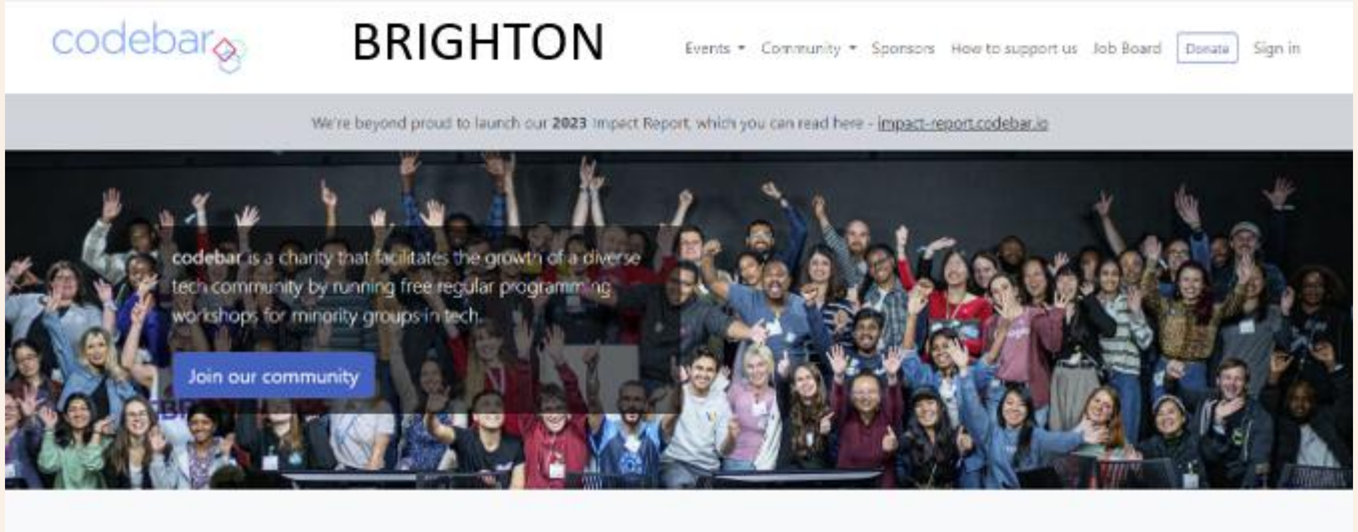

Brighton
I live ~ one mile


Leo - Fox Red Lab


Local Fox follows us home




Born/Lived in Frognal
ARCHWAY
FINSBURY PARK
Premier Inn London Hampstead hotel
43 min
6.5 miles
ALDI
ISLINGTON
Shurgard Self Storage London Euston
41 min
6.9 miles
PADDINGTON
WHITECHAP
EC4R 3AD
Hyde Park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We're beyond proud to launch our 2023 Impact Report, which you can read here - impact-report.codebar.io
codebar is a charity that facilitates the growth of a diverse tech community by running free regular programming workshops for minority groups in tech.
Join our community

Why Evals?

Ensure TRUST and CAPABILITIES of our AGENTS
so that somebody will pay for them.

Would you pay for a substandard product?

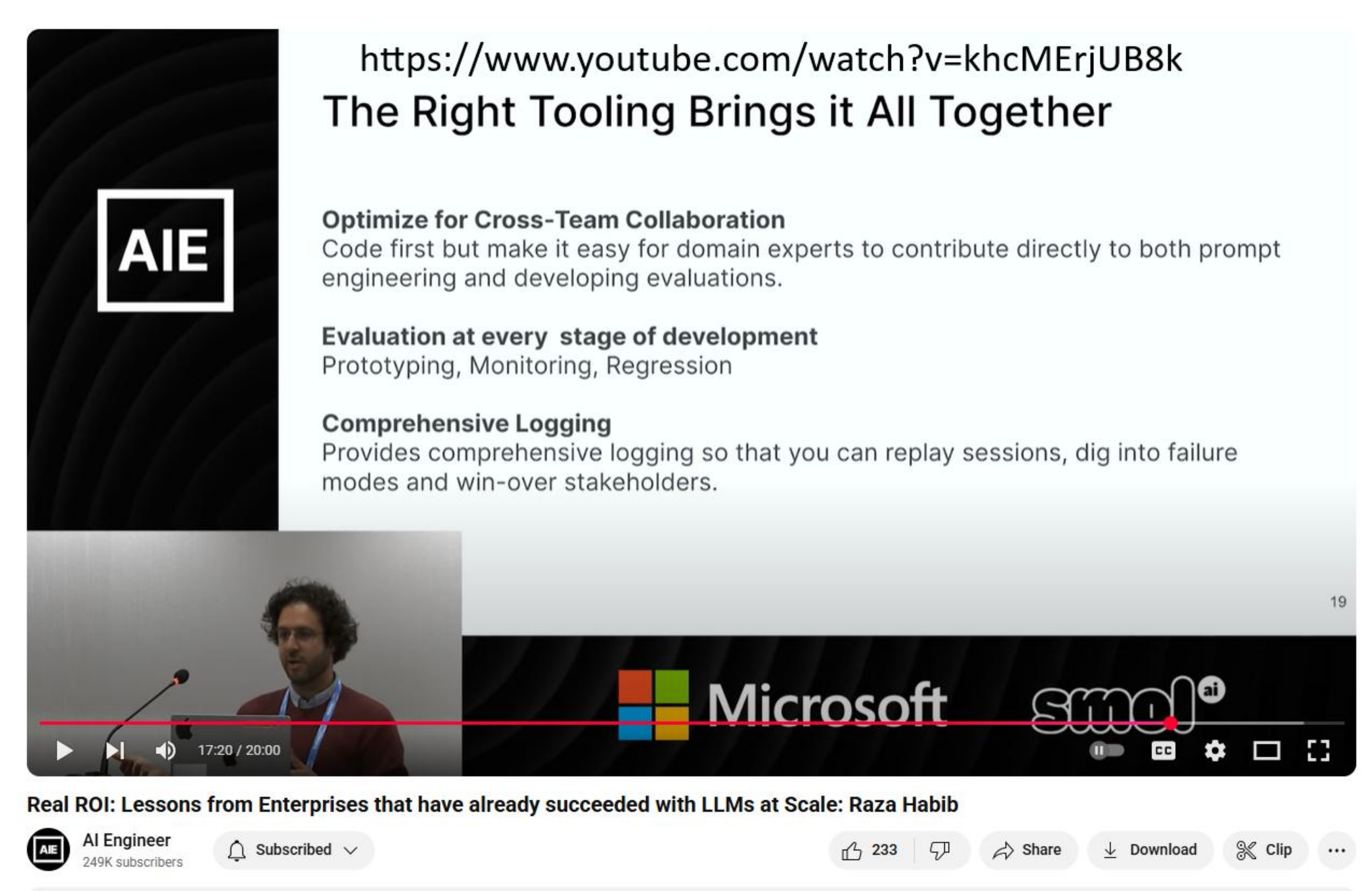
Excellence in an evaluator will rub off on developer prompt code.

Can we save £ if we use a different LLM?

Model costs

Cost of a unit of ‘intelligence has dropped ~ 99% in last two years and is 10x a year (anecdotal)

When to use Evals?



Raza Habib is the CEO and Cofounder of Humanloop

Acquired by Anthropic

What is an AI Agent?

Many definitions and people opt for ‘Agentic Apps’

Anthropic

- **Workflows** are systems where LLMs and tools are orchestrated through predefined code paths.
- **Agents**, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

API

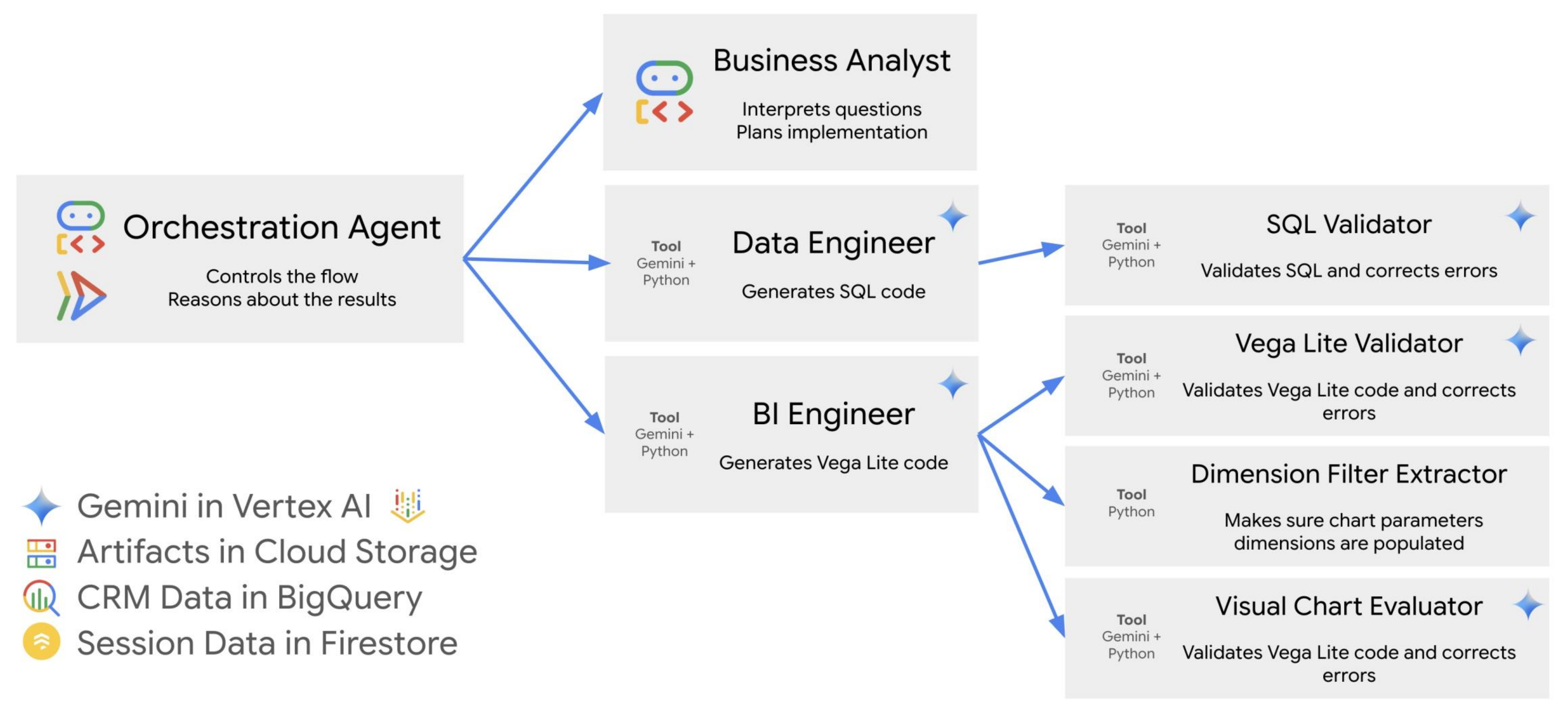
```
model = "gpt-4o-mini"  # Model
model_endpoint = "https://api.openai.com/v1/chat/completions"  # just one endpoint
headers = {
    "Content-Type": "application/json",  # Authorisation
    "Authorization": f"Bearer {api_key}",
}
# payload structure may vary from LLM Organisation but it is a text string.
payload = {
    "model": model,
    "messages": [
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_prompt},
    ],
    # additional parameters
    "stream": False,
    "temperature": temperature,
}
# Use HTTP POST method
response = requests.post(
    url=model_endpoint,  # The endpoint we are sending the request to. Low Temperature:
    headers=headers,  # Headers for authentication etc
    data=json.dumps(payload),  # Inputs, Context, Instructions, Additonal parameters
).json()
```

These messages contain Instructions and Context

But I don’t use Agents!

- Perhaps not now, but it might be that your app is required to use Natural Language, (text, voice) from the user.
- You might use or incorporate ready made 3rd party software that is Agentic as in the next slide…Google’s CRM Agent…
- or use Model Context Protocol/Agent2Agent in your ‘app’
- It may be helpful to understand what Agents are

Google CRM Agent

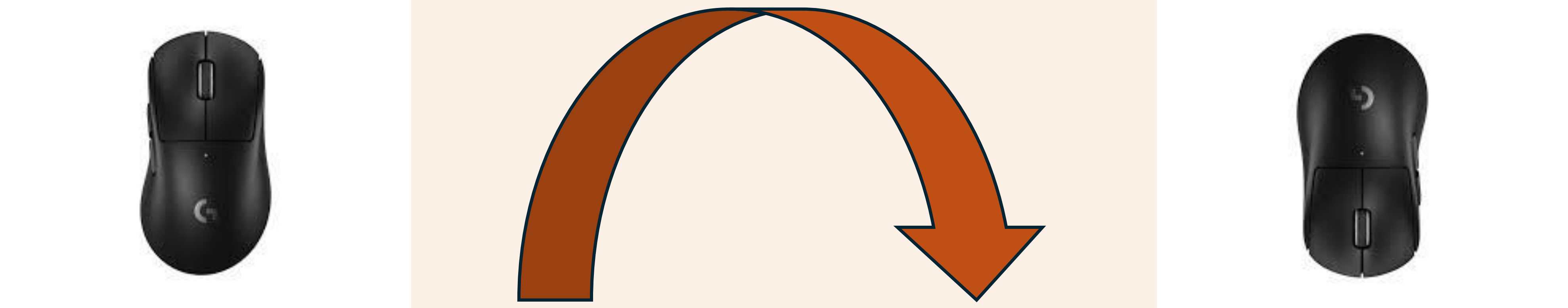


https://github.com/vladkol/crm-data-agent/blob/main/src/agents/data_agent/prompts/crm_business_analyst.py
https://www.youtube.com/watch?v=2-AJszogj7Y

What is Prompt/Context Engineering?

- Prompt Engineering is going out of fashion and people opt for *Context Engineering* and *In Context Learning*.
- We create a set of instructions and add **additional information** to a request we send to the LLM.
- We can let LLM know that more context is available if it tells us which of the **tools/functions** we have told it about it wants us to call with **arguments** it has provided – we will run it on our box and send the extra context/information.
- The API is STATELESS – we must pass all the information it needs every time as if it was the first time.

180-degree shift



No more difficult than what we do currently – just 180 degrees different

“It doesn’t get any easier – just different” - Anon

Where to start? What to do?

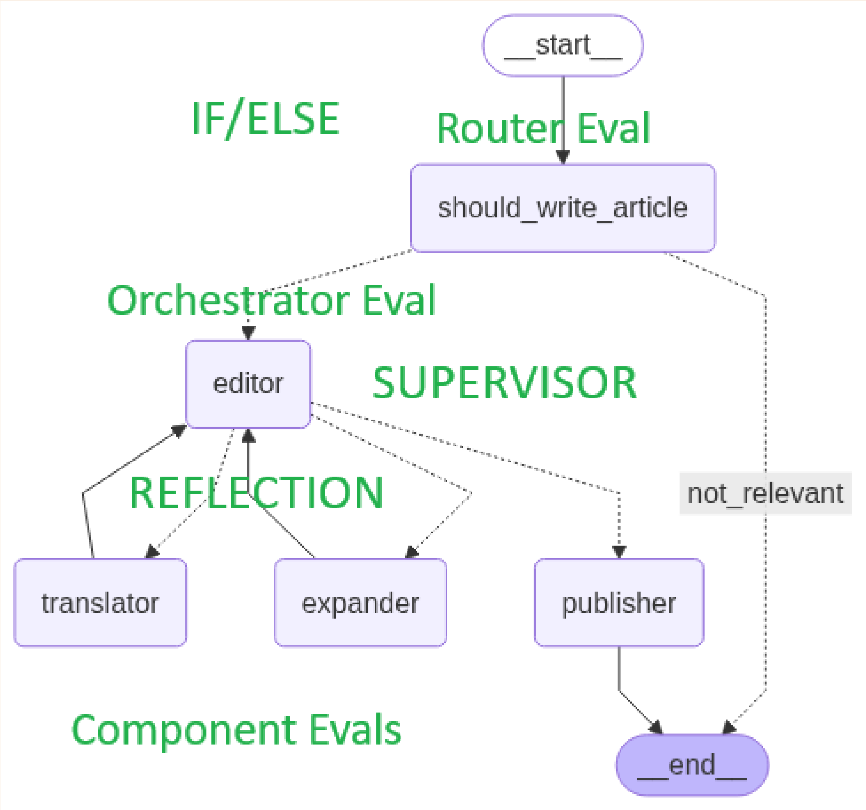
- We will look at Case Study 1 in the manual and framework.

Given an article title, determine if it is about AI.

1. Take title and create article of N words (EXPANDER)
2. In a specified language (TRANSLATOR)
3. Ensure article is not sensational (EDITOR)

PUBLISH = YES only if all three are YES

This app could be a leaf of a larger app.



A possible workflow

We build our app with a basic prompt “Write an article on {topic} of around {N} words.”

It all works mechanistically.

Now lets progress…

A possible workflow

We decide to add a simple 💛

We notice that users like:

Bullet points – Abstract - Conclusion

A possible workflow

We change our prompt to include these features.

However, only 2% of users give feedback so we decide to use Human Evaluators to give feedback as well as label articles ‘good’ etc

A possible workflow

We need to draft up guidelines on what an article should contain, its structure and other ways an article is ‘good’.

This exercise further helps our LLM app prompt, and we refine it.

A possible workflow

We can only handle/afford a small percentage, so we decide to use LLM as Judge.

Seems illogical but is actually very effective.

With our original prompt and our human evaluator guidelines we write a prompt for this LLM Judge. We can now process a much greater percentage online and offline.

It is better than our app! We refine our app prompt…

A possible workflow

Now that we are looking at our data, we see that sometimes the app offers financial advice.

We must prevent this (guardrails) so we create an LLM as Judge that detects if this article had financial advice.

It saves many problems…

A possible workflow

Now that we have a robust way of evaluating our app, we can start to tinker:

- Different models in different places for cost reduction yet same performance?
- Does one part of the app give greater benefit for unit of change than another?
- Can we reduce latency without breaking anything (regression testing)
- Offer a client dashboard of filtered metrics to our clients in real time? They have entrusted us with their business.
- *Process not tools.*

Demo – our app from earlier

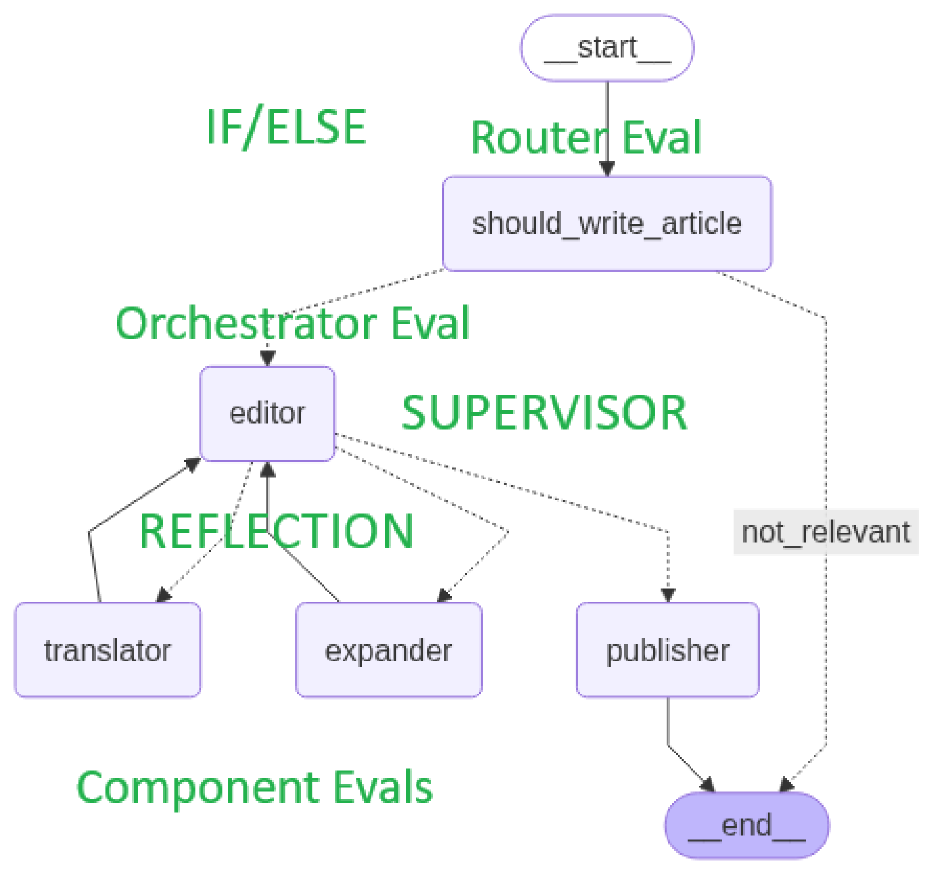
We will look at Case Study 1 in the manual and framework.

- Given an article title, determine if it is about AI (ROUTER)
- Take title and create article of N words (EXPANDER)
- In a specified language (TRANSLATOR)
- Ensure article is not sensational (EDITOR)

A ‘good article’ – OMITTED IN DEMO

PUBLISH = YES only if last three are YES in EDITOR

Let’s go to the guide and code…



Summary Key Points

- We need to look at the data to determine what are EVAL criteria are
- We need to determine what success will look like
- We will need humans at certain points
- We need a Domain Expert
- LLM Judges are primarily for scaling
- We need to do EVALS on these LLM Judges
- Process more important than tools

Future

Craig West

https://craig-west.netlify.app/

https://evaluating-ai-agents.com/

Contact links on the sites

Always have 10 mins to answer questions...